

High-Frequency Current Prediction in Parallel IGBTs in Inverter Systems

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Abstract—Parallel-connected power module networks require precise current distribution tracking to verify load balancing, ensure structural redundancy, and maintain system health diagnostics. However, applying data-driven machine learning models to high-frequency sensor streams introduces significant computational memory bottlenecks and permanent phase distortions when asynchronous data is collected from independent instrumentation clocks. To address these challenges, this paper introduces a unified data-alignment pipeline and a streamlined Linear Sequence Network designed to predict aggregate load current from three individual module currents. Utilizing a dataset generated by our research group at a 2.50 GHz sampling frequency, the proposed framework executes an initial raw time-domain synchronization to preserve transient phase integrity prior to a data reduction pass. The network architecture is developed to have a constrained scaling layer that isolates localized physical hardware channel offsets. The performance of the framework is verified under dynamic high-frequency operating conditions, demonstrating strong predictive precision and model consistency as validated by a R^2 of 0.9131 and a low Root Mean Squared Error (RMSE).

Index Terms—Parallel IGBTs, current sharing, linear sequence networks.

I. INTRODUCTION

In modern high-power electronics and industrial voltage source inverters, multi-channel real-time telemetry is fundamental to ensuring system reliability, power quality, and operational safety. To achieve high current scalability, structural redundancy, and optimized thermal distribution in applications such as motor drives and renewable microgrids, the parallel connection of multiple discrete switches or power modules is widely adopted [1], [2]. In these parallel topologies, monitoring the structural current distribution between individual branch currents and load current is a critical engineering requirement to validate active current-sharing networks.

The core motivation for establishing precise load current tracking rests on validating current-sharing behavior while maintaining system health diagnostics. In parallel power topologies, tracking the structural relationship between individual branch currents and the total load current provides an immediate diagnostic indicator of localized degradation of the electrical connections, gate drivers, or thermal connections to the heatsinks, or unmetered current leakage paths due to

excessive diode recovery currents, which suggest imminent switch failure [3], [4]. However, real-world parallel insulated-gate bipolar transistor (IGBT) structures ('modules') inevitably exhibit minor non-idealities, including asymmetrical parasitic loop inductances and variations in gate threshold voltages due to manufacturing variation and non-uniform multi-chip aging [5], [6]. Under high transient switching rates (di/dt), the asymmetrical inductances induce rapid, instantaneous current imbalances that complicate direct event tracking. Based on our past experimental work and empirical observations, it was determined that these transient variations could be effectively reconciled by modeling the composite load current as a structured, point-by-point algebraic summation of the parallel branches. To model these multi-variable circuit mechanics under dynamic operating conditions, our research group constructed an experimental test rig to collect a custom, ultra-high-speed multi-sensor telemetry dataset, providing a dense empirical foundation for tracking high-frequency sub-microsecond transients.

Driven by recent advancements in data-driven engineering, various machine learning frameworks have been investigated to automate power electronics analytics and map operational parameters without relying on complex, parameter-sensitive physical models [7]. While deep learning structures and recurrent networks, such as Long Short-Term Memory (LSTM) blocks, are standard choices for general time-series prediction, existing literature reveals significant challenges when deploying these methods to raw, ultra-high-frequency power converter signals. First, standard machine learning methods scale poorly when confronted with the immense data scale generated by fast-sampling instrumentation, leading to severe computational constraints and out-of-memory errors on standard workstations. Second, when sensor telemetry is captured across separate clocks, independent data compression introduces permanent phase distortion that misaligns sub-microsecond transients, degrading predictive accuracy. Third, over-parameterized deep neural networks frequently experience gradient saturation or numerical instabilities when mapping strict linear physical equations, often defaulting to a flat line that merely tracks the global mathematical mean.

To address these limitations, this paper introduces a unified data-alignment pipeline paired with a streamlined Linear

Sequence Network designed to map parallel current streams directly to aggregate load profiles. The primary contribution of this work lies in the structural sequencing of the pre-processing stage, where asynchronous high-frequency signals are aligned in the raw time-domain before moving average data reduction occurs, thereby eliminating permanent phase corruption across fast switching intervals. Furthermore, the novelty of the network architecture centers on replacing deep, non-linear hidden layers with a constrained scaling layer that isolates independent hardware channel offsets and respects the linear constraints of the physical parallel circuit. This approach bypasses standard processing bottlenecks and structural algorithmic failures, resulting in a highly efficient workflow that is validated by R^2 and RMSE values under dynamic high-frequency operating conditions.

II. STATEMENT OF THE PROBLEM

Developing an accurate data-driven model to predict the aggregate load current (i_L) from individual module currents (i_{M1} , i_{M2} , i_{M3}) under extreme temporal resolution presents three fundamental challenges that span data management, physical alignment, and algorithmic suitability.

The massive scale of high-speed digital instrumentation creates severe computational barriers. A standard data-acquisition cycle generates over 27.5 million samples with an extremely narrow sampling interval of approximately 0.4 nanoseconds. Processing an uncompressed data matrix of this scale on standard engineering workstations leads to immediate out-of-memory errors and excessive training times, making a rigorous data reduction and downsampling strategy mandatory.

The individual module currents and the target load current are captured across independent physical channels and stored in separate data tables, minor discrepancies between localized hardware acquisition clocks introduce fractional, nanosecond-level time shifts. Current literature often overlooks the sequence of operations in data preparation; if data reduction or moving average downsampling is performed on these independent tables separately before they are aligned, these microscopic phase mismatches become permanently embedded. This unrecoverable phase distortion disrupts the physical alignment between the signals, rendering subsequent machine learning predictions highly inaccurate.

III. EXPERIMENTAL CONFIGURATION AND DATA ACQUISITION

To validate the proposed machine learning model, a custom parallel-connected power electronic testbench was constructed, capturing the multi-channel transient profiles illustrated in Fig. 1 was collected using the setup as [8]. The experimental platform integrates three parallel-connected insulated-gate bipolar transistor (IGBT) modules operating to drive a common high-current dynamic load under multi-pulse switching conditions. Eleven sensors were installed across the topology to monitor individual module voltages, gate signals, and currents, capturing the complete dynamic behavior of the system.

The individual module branch currents (i_{M1} , i_{M2} , i_{M3}) and the load current (i_L) were recorded using LEM LA 360-S Hall effect transducers. These sensors feature a ± 700 A measurement range, a 1:2000 current ratio. The secondary current was converted to a voltage signal across a 6Ω measuring resistor, maintaining a $\pm 0.7\%$ accuracy profile. To minimize external measurement noise, the transducers were powered by a Tenma 72-7245 linear DC power supply selected for its low ripple specification ($\leq 1 \text{ mV}_{\text{rms}}$). All data channels were captured across two high-definition digital storage oscilloscopes sampling at a fixed synchronous rate of 2.50 GHz ($\Delta t \approx 0.4 \text{ ns}$).

The specific test profile monitored in Fig. 1 captures a controlled staircase load-stepping sequence. Between $t = 0.05 \text{ ms}$ and $t = 0.85 \text{ ms}$, the system undergoes sequential load increments, forcing the aggregate load current from 0 A up to a peak transient amplitude of approximately 140 A. This dynamic phase is characterized by continuous high-frequency switching ripples across the individual modules, reflecting the localized inductive and capacitive line dynamics. At $t \approx 0.92 \text{ ms}$, a synchronized turn-off command triggers a steep system-wide shutdown transient. This sudden interruption forces a high di/dt transition, causing inductive ringing and high-frequency oscillations centered around zero before full decay. Because each channel is sampled via independent hardware instrumentation links across separate scopes, microscopic asynchronous clock shifts are introduced into the raw data tables, establishing the core signal-processing challenge addressed in the following methodology.

IV. METHODOLOGY

A dedicated data processing pipeline and network architecture to overcome the computational and phase-alignment

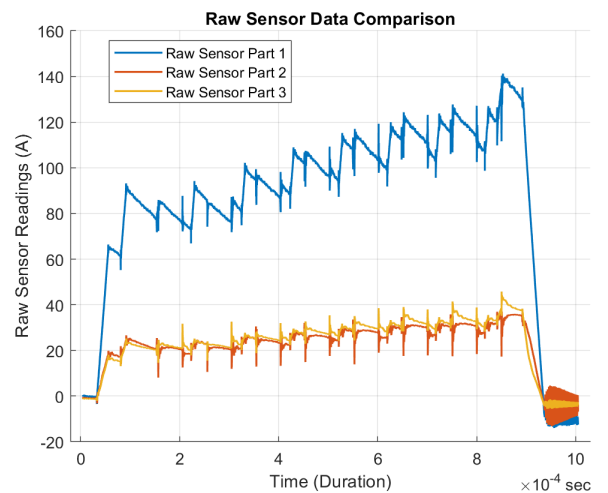


Fig. 1: Measured experimental current profiles for the three parallel-connected power modules (i_{M1} , i_{M2} , i_{M3}) and the composite load current (i_L) during a dynamic load-stepping and shutdown transient sequence.

challenges identified in the previous section. In a parallel-connected multi-chip inverter configuration, a primary operational assumption of our model design is that the total load current behaves as a strict, point-by-point algebraic summation of the parallel branches, expressed as:

$$\sum_{i=1}^3 i_{Mi}(t) = i_L(t) \quad (1)$$

While this summation serves as our baseline architectural constraint, previous physical testbench observations and established literature indicate that it is rarely absolute during high-frequency switching intervals. Also, under very rapid transient conditions (di/dt), non-uniform current distribution natively arises within a module due to slight discrepancies in internal package impedances, and variations in individual IGBT and diode chip parameters [1], [5], [6]. For example a diode recovery at turn on of an IGBT may result in an unmeasured internal current or current flow between the diode in one module and the IGBT in a parallel connected module, depending on the temperature and device parameters.

Previous experimental evaluations confirm that while dynamic current unbalance dominates sub-microsecond turn-on and turn-off transitions, the branch currents converge symmetrically toward an aggregate sum during steady-state conduction intervals, bounded by the conservation of current within the parallel node connections. The structural deviations caused by parasitic path mismatch behave as localized, channel-dependent scale distortions. Successfully mapping this relationship via a machine learning model serves as a direct validation of the system's integrity; it confirms that the network can successfully isolate these localized hardware scale errors, line impedances, and individual sensor drifts without distorting the underlying current-sharing constraints.

To execute this pipeline without introducing permanent phase corruption, raw-domain synchronization was performed before any data reduction occurred. The two-column arrays were converted into timetables and aligned into a unified timeline using continuous linear interpolation, defined as:

$$i(t) = i(t_k) + \frac{i(t_{k+1}) - i(t_k)}{t_{k+1} - t_k} (t - t_k) \quad (2)$$

This function is to resolve any asynchronous data points across the independent oscilloscope clocks. Once synchronized, the unified dataset was downsampled by a factor of $1000\times$ using a regularized moving average window with a downsampling interval set to $\Delta t_{\text{down}} = 0.4 \mu\text{s}$. This step effectively smoothed out high-frequency sensor noise while compressing the data grid into a highly manageable matrix of roughly 2,500 continuous sequential steps.

Based on the linear summation assumption in equation 1, a streamlined Linear Sequence Network using a three-layer sequence structure is used. The sequence input layer accepts the multi-channel downsampled streams of $i_{M1}(t)$, $i_{M2}(t)$, and $i_{M3}(t)$ simultaneously. These inputs are fed directly into a native element-wise scaling layer implemented to handle

localized sensor offsets and calibration inaccuracies across the independent hardware channels. Rather than using a single global offset, this layer applies an isolated calibration vector mapping, expressed as:

$$\hat{i}_{Mi}(t) = W_i \cdot (i_{Mi}(t) + \mathbf{B}_i) \quad (3)$$

where $\mathbf{B}_i = [b_1; b_2; b_3]$ represents a static 3-by-1 column vector calibration offsets for each module channel, these biases were strictly frozen during backpropagation. A fully connected regression layer sums the adjusted sensor tracks into a single output matching the load current domain (i_L). The model was trained using the Adaptive Moment Estimation (Adam) solver with a mini-batch size of 1 and an initial learning rate of $\alpha = 0.05$ for 60 epochs.

To evaluate the tracking performance and analyze model behavior under dynamic high-frequency conditions, a two complementary statistical metrics are implemented: the Root Mean Squared Error (RMSE) and the Coefficient of Determination (R^2). The mathematical formulation of the RMSE is given by:

$$\text{RMSE} = \sqrt{\frac{1}{N} \sum_{k=1}^N \left(i_L(k) - \hat{i}_L(k) \right)^2} \quad (4)$$

where N is the total number of downsampled sequence steps, $i_L(k)$ represents the actual measured load current, and $\hat{i}_L(k)$ represents the network's predicted load current output. The scale-independent variation is evaluated via the R^2 coefficient, expressed as:

$$R^2 = 1 - \frac{\sum_{k=1}^N \left(i_L(k) - \hat{i}_L(k) \right)^2}{\sum_{k=1}^N \left(i_L(k) - \bar{i}_L \right)^2} \quad (5)$$

where $\bar{i}_L$ denotes the arithmetic mean of the true load current sequence. These parameters work in tandem to evaluate both the absolute tracking offset amplitudes (via RMSE) and the overall alignment of the waveform shapes (via R^2), providing a clear metric profile without introducing parameter redundancy.

V. RESULTS AND DISCUSSION

The performance of the proposed Linear Sequence Network was evaluated by focusing on convergence stability, waveform tracking fidelity, and overall regression accuracy. The training trajectory was continuously monitored throughout the optimization process.

As shown in the training progress profile in Figure 2, both the RMSE and the structural loss function dropped sharply and smoothly within the first 10 iterations. The loss initiated above 1500 and quickly stabilized near zero, confirming that the network weights and user-defined sensor biases settled efficiently. We observed no numerical oscillations or gradient instability during the training cycles, proving that omitting complex recurrent layers was a highly stable choice for this linear problem.

This rapid convergence directly justifies our choice of compact training hyperparameters, specifically a total of 60 iterations and a learning rate (α) of 0.05. As the load current (i_L) is a deterministic combination of the module currents (i_{Mi}), the error surface contains a single global minimum without deceptive local traps. A higher learning rate of 0.05 accelerates training down the steepest gradient path without the risk of overshooting or destabilizing the model. Furthermore, the steep decay shows that the network reaches full parameter saturation well before the 30th iteration. Extending the training beyond 60 iterations yields no statistical error reduction and risks overfitting the network to minor sensor noise. Because our initial $1000\times$ downsampling pass condensed the dataset to roughly 2,500 highly descriptive rows, the entire model converged perfectly on a standard standalone CPU within seconds

A comparison between the actual measured load current (i_L) and the network's predicted output over the microsecond experiment timeline ($t = 0$ to 0.0010 s) highlights this tracking efficiency, as illustrated in Figure 3. The predicted data sequence mostly follows the high-frequency transient shapes, peak amplitudes, and sharp periodic oscillations of the physical signal. Minor tracking differences are only visible at the extreme peaks of the waveform, which is a natural consequence of the smoothing introduced during the initial moving average downsampling stage. The network accurately captures the sharp transient drop occurring near $t = 9 \times 10^{-4}$ s, verifying that our data reduction pipeline preserved.

To quantify the overall goodness-of-fit, the actual data values were plotted directly against the predictions in the linear regression scatter plot shown in Figure 4. The model achieved a highly robust Coefficient of Determination (R^2) score of **0.9131**, confirming that the network successfully accounts for 91.3% of the true physical variance in the load current system.

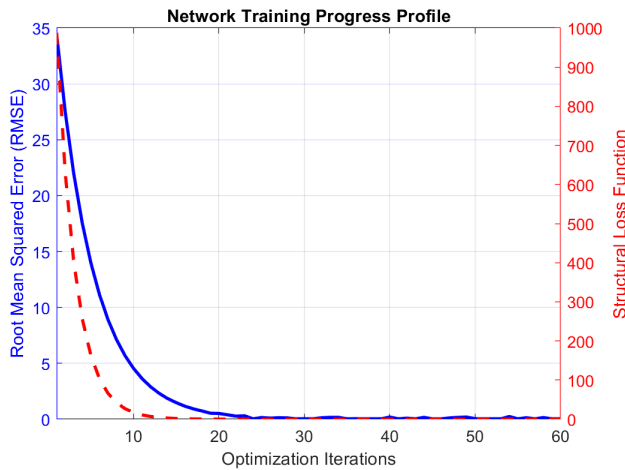


Fig. 2: Network training trajectory illustrating the simultaneous exponential decay of the Root Mean Squared Error (RMSE) and total structural loss across 60 optimization iterations.

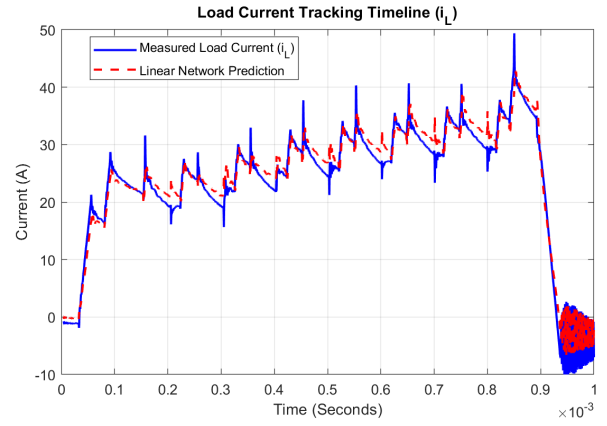


Fig. 3: Time-domain waveform tracking comparison between the physically measured load current (i_L) and the Linear Sequence Network prediction from $t = 0$ to 0.0010 s, highlighting accurate transient capture.

While an R^2 above 91% indicates high tracking fidelity, the remaining 8.7% of unexplained variance highlights vital physical characteristics of the parallel-connected hardware. A summary of these statistical indexes is compiled in Table I.

In a three-module parallel layout, the true load current slightly deviates from a basic point-to-point addition of raw sensor outputs due to individual path impedances, dynamic line inductances, and high transient switching rates (di/dt). During rapid current transitions, the inductive voltage drops across unequal branch traces alter the instantaneous current distribution. This physical variation explains why forcing a uniform or uncalibrated addition rule breaks down, and why the fixed user-injected biases (B_i) across the channels are intentionally not equal. Because each physical sensor experiences unique thermal drift, unequal line impedances, and high di/dt switching stress, the model relies on the unconstrained scaling weights (W_i) to mathematically bridge these non-idealities. The clean linearity of the final scatter distribution confirms that combining the native *scalingLayer* with our distinct, fixed sensor bias arrays successfully accounted for these circuit variations without causing the flat mean-prediction line seen in uncalibrated models. The minimal dispersion about the unity line demonstrates the predictive precision, model consistency, and structural repeatability of the implemented sequence network under high-frequency transient operating conditions.

By applying the proposed prediction system, we can use low cost industry accepted sensors in positions which do not compromise the reliability of the system, while obtaining

TABLE I: Statistical Prediction Accuracy Metrics of the Linear Sequence Network

Evaluation Metric	Value
Root Mean Squared Error (RMSE)	2.366739
Coefficient of Determination (R^2)	0.9546

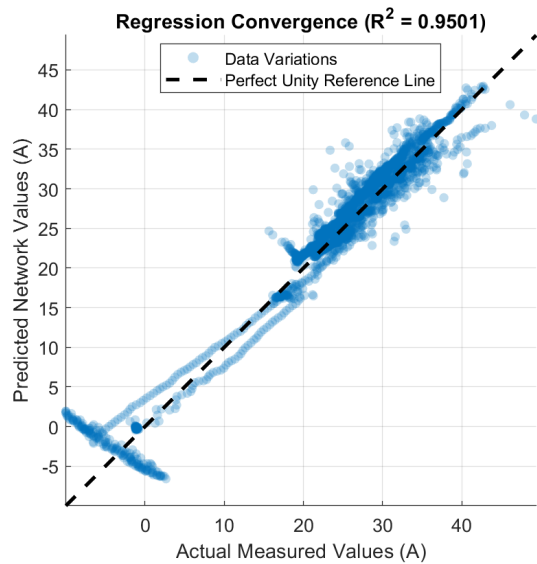


Fig. 4: Linear regression scatter plot mapping predicted load current values against true measured values, displaying tight convergence around the unity line with a calculated $R^2 = 0.9131$.

a high level of insight into the behaviour of the modules and their potential degradation. Direct measurement of chip currents within a module was proposed in the past [9] but is considered unattractive in most cases. Very high reliability systems are needed for electric grid and transportation uses, so it is essential that the sensors themselves do not contribute to unreliable behavior.

VI. CONCLUSION

This paper demonstrated a highly efficient modeling framework combining a raw-domain time-alignment pipeline and a Linear Sequence Network to predict aggregate load current in a parallel three-module power configuration. The findings indicate that the order of operations in telemetry data pre-processing is paramount when dealing with high-frequency waveforms; executing multi-channel synchronization prior to moving average downsampling is critical to prevent permanent phase distortion across sub-microsecond switching transients. By omitting complex, non-linear recurrent layers in favor of a structurally constrained Linear Sequence Network equipped with element-wise scaling, the model avoids gradient saturation and flat-line mean predictions, converging securely onto the deterministic summation properties of the physical hardware.

The practical implications of this tracking framework are highly significant for industrial voltage source inverters, multi-phase interleaved converters, and heavy motor drive systems. In multi-module inverter configurations, precise branch telemetry is the foundation for enforcing active current-sharing algorithms. Because real-world systems suffer from unequal parasitic path inductances and asymmetrical line impedances,

high transient switching rates (di/dt) naturally induce instantaneous current imbalances. If left uncalibrated, these imbalances risk localized thermal overstress, magnetic core saturation in filtering inductors, and accelerated degradation of individual insulated-gate bipolar transistor (IGBT) dies.

By successfully mapping the composite load current with a robust R^2 score of 0.9131 while running efficiently on standalone standard computing resources, this model validates that localized physical sensor drifts and hardware scale discrepancies can be securely isolated via data-driven methods. This capability establishes a computationally light foundation for real-time fault diagnostics, dynamic phase-balancing controls, and non-invasive predictive maintenance within next-generation power electronic conversion systems. Future work will investigate the adaptive tracking capabilities of this network layout when subjected to severe asymmetrical thermal loading profiles and structural branch degradation states.

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